

XINHAI HOU

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EDUCATION

University of Michigan, Ann Arbor, U.S.

Ph.D. in Bioinformatics and Scientific Computing | GPA: 4.0

Sep 2021 - May 2026 (expected)

- Advisor: **Prof. Todd Hollon**, **Prof. Honglak Lee**
- Research interests: vision language model, reinforcement learning, agentic system
- Published paper in **Nature** (FastGlioma), **CVPR** (CodeV, HiDisc, highlight top 2.5%), **NeurIPS** (OpenSRH)
- Services: conference reviewer for ICLR 2025, NeurIPS 2022-2025, ECCV 2024, CVPR 2024-2026

The Chinese University of Hong Kong, Shenzhen, China

B.S. in Statistical Science, First Class Honours (top 10%)

Sep 2016 - May 2020

PROFESSIONAL EXPERIENCE

Applied Scientist Intern | **Amazon** (Seattle)

May 2025 - Aug 2025

- Led an end-to-end research project CodeV, addressing production bottlenecks in infographic extraction and multi-object counting/grounding, resulting in a paper accepted to **CVPR 2026**.
- Post-train Qwen2.5-VL-7B for agentic visual reasoning on interleaved text, Python code, image with GRPO, boosting performance (by average 3.2%) and achieve SOTA on 10 visual perception and reasoning benchmarks.
- Identify and mitigate unfaithful tool usage by rubric-based reward and expert SFT pipeline (increase 30% faithfulness).

Research Assistant | **University of Michigan**

Sep 2021 - present

- Developed FastGlioma, a visual foundation model for fast (10x less) tumor detection. (**Nature, 2025**)
- Proposed Slide Pretraining Transformer (SPT) for gigapixel slide image embeddings. (**NeurIPS 2024, AIM-FM**)
- Introduced HiDisc, a hierarchical self-supervised learning framework for microscopy images. (**CVPR highlight, 2023**)
- Curated and released the first Stimulated Raman Histology dataset and benchmark. (**NeurIPS D&B track, 2022**)
- Built an LLM agent for patient survival forecasting by constructing an evaluation pipeline from 10M+ medical reports, improving the C-index by 15% via a multi-agent system with self-calibration mechanisms.

Machine Learning Engineer | **Tencent** (Beijing), advised by Dr. **Pengfei Xiong**

Feb 2021 - Aug 2021

- Fine-tuned GPT-2 and UniLM on social media and news corpus for title generation used in WeChat Search.
- Doubled the training speed by distributed training and increased 5% in the ROUGE score by beam search.
- Won **fifth** award out of 4335 participants in an ACM challenge: [Multimodal Video Advertisement Competition](#).

SELECTED PUBLICATIONS

(* for co-first contribution)

[CodeV: Code with Images for Faithful Visual Reasoning via Tool-Aware Policy Optimization](#)

Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026

Xinhai Hou, ..., Todd C. Hollon, Bryan Wang.

[Foundation models for fast, label-free detection of glioma infiltration](#)

Nature, 2025

Akhil Kondepudi, Melike Pekmezci, **Xinhai Hou**, ..., Todd Hollon.

[A self-supervised framework for learning whole slide representations](#)

Neural Information Processing Systems (**NeurIPS**) Workshop on AIM-FM, 2024

Xinhai Hou, ..., Honglak Lee, and Todd C. Hollon.

Hierarchical discriminative learning improves visual representations of biomedical microscopy

Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023 (Highlight, top 2.5%)

Cheng Jiang*, **Xinhai Hou***, ..., Honglak Lee, and Todd C. Hollon.

OpenSRH: optimizing brain tumor surgery using intraoperative stimulated Raman histology

Conference on Neural Information Processing Systems (**NeurIPS**) Datasets and Benchmarks Track, 2022

Cheng Jiang*, Asadur Chowdury*, **Xinhai Hou***, ..., Honglak Lee, and Todd C. Hollon.

OTHER PUBLICATIONS

Step-calibrated diffusion for biomedical optical image restoration

AAAI Conference on Artificial Intelligence, 2025

Yiwei Lyu,..., Xinhai Hou, ..., Honglak Lee, and Todd C. Hollon

Super-resolution of biomedical volumes with 2D supervision

CVPR Workshop on Computer Vision for Microscopy Image Analysis, 2024

Cheng Jiang, ... Xinhai Hou, ..., Honglak Lee, and Todd C. Hollon

Valproic acid-induced changes of 4D nuclear morphology in astrocyte cells

Molecular biology of the cell, 2021

Alexandr Kalinin, Xinhai Hou, ..., Brian Athey.

CORE COURSEWORK

Machine Learning	Computer Vision, Large Language Modeling, Machine Learning foundations
Statistics	Data Mining, Statistical Inference, Time Series, Nonparametric Statistics
Computer Science	Data Structure and Algorithm, Parallel Computing, Database Management
Bioinformatics	Bioinformatics concepts and algorithms, Biology for computational scientists

SKILLS

Programming languages	Python, Shell, C/C++
Framework	uv, Pytorch, vLLM, VeRL, OpenCV
Tools	Git, Vim, VS Code, AWS, L ^A T _E X, Adobe Illustrator

AWARDS

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| • MICDE Graduate Fellowship
Michigan Institute for Computational Discovery and Engineering, U of M | 2024-2025 |
| • CVPR 2023 Scholarship | 2023 |
| • Dean's List (Top 5%)
The Chinese University of Hong Kong, Shenzhen | 2016 - 2020 |
| • Undergraduate Research Award
The Chinese University of Hong Kong, Shenzhen | 2016 - 2020 |